

Definition 1 For each non-negative integer n define a function ψ_n by

$$\psi_n(z) := \frac{z^n}{\sqrt{n!}}.$$

Denote by $\mathcal{H}$ the linear space of all entire functions of the form

$$f(z) := \sum_{n=0}^{\infty} c_n \cdot \psi_n(z) = \sum_{n=0}^{\infty} c_n \cdot \frac{z^n}{\sqrt{n!}}, \quad \sum_{n=0}^{\infty} |c_n|^2 < \infty.$$

Define a complex inner product $\langle g, f \rangle$, also denoted by $\langle g|f \rangle$, on $\mathcal{H}$ by

$$\langle g, f \rangle := \frac{1}{\pi} \cdot \iint \overline{g(z)} \cdot f(z) \cdot e^{-|z|^2} dx dy := \frac{1}{\pi} \cdot \int_{\rho=0}^{\rho=\infty} \int_{\theta=-\pi}^{\theta=\pi} \overline{g(\rho \cdot e^{i\theta})} \cdot f(\rho \cdot e^{i\theta}) \cdot e^{-\rho^2} d\theta \rho d\rho.$$

Theorem 2 The sequence $(\psi_n)_{n=0}^{\infty}$ forms an orthonormal basis of $\mathcal{H}$: with $c_n = \langle \psi_n | f \rangle$.

Proof. If $k = n = \ell$, then

$$\langle \psi_n, \psi_n \rangle = \frac{1}{\pi} \cdot \int_{\rho=0}^{\rho=\infty} \int_{\theta=-\pi}^{\theta=\pi} \frac{\rho^n \cdot e^{-n\theta}}{\sqrt{n!}} \cdot \frac{\rho^n \cdot e^{n\theta}}{\sqrt{n!}} \cdot e^{-\rho^2} d\theta \rho d\rho = \int_{u=0}^{u=\infty} \frac{u^n \cdot e^{-u}}{n!} du = 1.$$

If $k \neq \ell$, then

$$\langle \psi_\ell, \psi_k \rangle = \frac{1}{\pi} \cdot \int_{\rho=0}^{\rho=\infty} \int_{\theta=-\pi}^{\theta=\pi} \frac{\rho^n \cdot e^{-\ell\theta}}{\sqrt{\ell!}} \cdot \frac{\rho^n \cdot e^{k\theta}}{\sqrt{k!}} \cdot e^{-\rho^2} d\theta \rho d\rho = \int_{u=0}^{u=\infty} 0 du = 0.$$

Definition 3 The *identity* function $\iota : \mathbb{C} \rightarrow \mathbb{C}$ is defined for each $z \in \mathbb{C}$ by $\iota(z) := z$.

Denote differentiation by D , so that $Df = f'$ for each entire function f .

Define operators $a, a^*, P, Q, H : \mathcal{H} \rightarrow \mathcal{H}$ by $a := D$ and $a^* f := \iota \cdot f$ as well as

$$\begin{aligned} af &:= Df, & a^* f &:= \iota \cdot f \\ Q &:= \frac{1}{\sqrt{2 \cdot \omega}} \cdot (a + a^*), & P &:= \frac{\sqrt{\omega}}{i\sqrt{2}} \cdot (a - a^*) \\ H &:= \omega \cdot \left(a^* \circ a + \frac{1}{2} \right), & (Hf)(z) &:= \omega \cdot \left[z \cdot f'(z) + \frac{1}{2} \cdot f(z) \right] \end{aligned}$$

Theorem 4 Each function ψ_n is an eigenfunction of H with eigenvalue $\omega \cdot (n + \frac{1}{2})$:

$$H\psi_n = \omega \cdot \left(n + \frac{1}{2} \right) \cdot \psi_n, \quad H = \sum_{n=0}^{\infty} \omega \cdot \left(n + \frac{1}{2} \right) \cdot |\psi_n\rangle \langle \psi_n|.$$

Proof. It suffices to verify that $a^* a \psi_n = n \cdot \psi_n$:

$$\psi'_n(z) = n \cdot \frac{z^{n-1}}{\sqrt{n!}} = \sqrt{n} \cdot \frac{z^{n-1}}{\sqrt{(n-1)!}}, \quad z \cdot \psi'_n(z) = z \cdot n \cdot \frac{z^{n-1}}{\sqrt{n!}} = n \cdot \frac{z^n}{\sqrt{n!}} = n \cdot \psi_n(z).$$

Theorem 5 The operators a^* , P , Q have no eigenfunctions in $\mathcal{H}$, but a has uncountably many.

Proof. $af = Df = \lambda f$ for $f(z) = C \cdot \exp(\lambda \cdot z) = \sum_{n=0}^{\infty} \frac{\lambda^n}{\sqrt{n!}} \cdot \frac{z^n}{\sqrt{n!}}$, with $\sum_{n=0}^{\infty} \left| \frac{\lambda^n}{\sqrt{n!}} \right|^2 = e^{(\lambda^2)}$.

$$Pf = \lambda f, \quad \frac{\sqrt{\omega}}{i\sqrt{2}} \cdot (a - a^*)f = \lambda f, \quad \frac{\sqrt{\omega}}{i\sqrt{2}} \cdot (f' - \iota \cdot f) = \lambda f.$$

With $\mu := \frac{i\sqrt{2}}{\sqrt{\omega}} \cdot \lambda$, solve a linear differential equation, but $\overline{\exp(z^2/2)} \cdot \exp(z^2/2) = \exp(\Re(z^2))$.

$$\begin{aligned} f'(z) - (\mu + z) \cdot f(z) &= 0, \\ \exp\left(-\mu \cdot z - \frac{z^2}{2}\right) \cdot [f'(z) - (\mu + z) \cdot f(z)] &= 0, \\ \exp\left(-\mu \cdot z - \frac{z^2}{2}\right) \cdot f(z) &= C, \\ f(z) &= C \cdot \exp\left(\mu \cdot z + \frac{z^2}{2}\right) = C \cdot \exp(\mu \cdot z) \cdot \exp\left(\frac{z^2}{2}\right) \notin \mathcal{H}. \end{aligned}$$

Theorem 6 The commutators are

$$[a, a^*] := a \circ a^* - a^* \circ a = 1, \quad [Q, P] := Q \circ P - P \circ Q = i.$$

The anti-commutator $\{a, a^*\} := a \circ a^* + a^* \circ a$ is such that $\{a, a^*\} \psi_n = (2 \cdot n + 1) \cdot \psi_n$.

Proof. $[a, a^*]f = a \circ a^*f - a^* \circ af = (z \cdot f)' - z \cdot (f') = f + z \cdot (f') - z \cdot (f') = f = 1 \cdot f$.

$$[a + a^*, a - a^*]f = (a + a^*) \circ (a - a^*) - (a - a^*) \circ (a + a^*) = 2 \cdot (a^* \circ a - a \circ a^*) = -2 \cdot [a, a^*],$$

$$\{a, a^*\} \psi_n = (a \circ a^* + a^* \circ a) \psi_n = D(\iota \cdot \psi_n) + \iota \cdot D\psi_n = \psi_n + \iota \cdot \psi'_n + \iota \cdot D\psi'_n = (2 \cdot n + 1) \cdot \psi_n$$

Theorem 7

$$H = \frac{P \circ P + \omega^2 \cdot Q \circ Q}{2}$$

Proof.

$$\omega^2 \cdot Q \circ Q = \omega^2 \cdot \frac{1}{\sqrt{2 \cdot \omega}} \cdot (a + a^*) \circ \frac{1}{\sqrt{2 \cdot \omega}} \cdot (a + a^*) = \omega \cdot \frac{a \circ a + a^* \circ a + a \circ a^* + a^* \circ a^*}{2},$$

$$P \circ P = \frac{\sqrt{\omega}}{i\sqrt{2}} \cdot (a - a^*) \circ \frac{\sqrt{\omega}}{i\sqrt{2}} \cdot (a - a^*) = -\omega \cdot \frac{a \circ a - a^* \circ a - a \circ a^* + a^* \circ a^*}{2}.$$

Consequently,

$$\begin{aligned} \omega^2 \cdot Q \circ Q + P \circ P &= \omega \cdot (a^* \circ a + a \circ a^*) = \omega \cdot \{a, a^*\}, \\ \frac{\omega^2 \cdot Q \circ Q + P \circ P}{2} \psi_n &= \omega \cdot \frac{\{a, a^*\}}{2} \psi_n = \omega \cdot \left(\frac{2 \cdot n + 1}{2} \right) \psi_n = \omega \cdot \left(n + \frac{1}{2} \right) \psi_n = H \psi_n \end{aligned}$$

Therefore, $\frac{\omega^2 \cdot Q \circ Q + P \circ P}{2} = H$ because the two linear operators agree on every basis function.

Definition 8 The *annihilation* operator is a . The *creation* operator is a^* .

Definition 9 A *Hilbert space* is a Cauchy-complete real or complex linear space $\mathcal{H}$ with an inner product $\langle \cdot, \cdot \rangle$ and a complete countable orthonormal basis (every Cauchy sequence converges).

Definition 10 With an inner product $\langle \cdot, \cdot \rangle$ on a Hilbert space $\mathcal{H}$, the *adjoint* (operator) of a linear operator $A : \mathcal{H} \rightarrow \mathcal{H}$ is a linear operator $A^* : \mathcal{H} \rightarrow \mathcal{H}$ with $\langle g, Af \rangle = \langle A^*g, f \rangle$ for all $f, g \in \mathcal{H}$.
An operator A is *self-adjoint* if and only if $A = A^*$.

Remark 11 In Definition 10, the operators A and A^* must be defined on all of $\mathcal{H}$.

Theorem 12 $(A \circ B)^* = B^* \circ A^*$ and $A^{**} = A$ as well as $(A \circ A^*)^* = A \circ A^*$.

Proof. Recall that $\langle g, f \rangle = \overline{\langle f, g \rangle}$ with complex inner products.

$$\begin{aligned} \langle g, ABf \rangle &= \langle A^*g, Bf \rangle = \langle B^*A^*g, f \rangle, & (AB)^* &= B^*A^*; \\ \langle g, A^*f \rangle &= \overline{\langle A^*f, g \rangle} = \overline{\langle f, Ag \rangle} = \langle Ag, f \rangle, & A^{**} &= A; \\ \langle g, AA^*f \rangle &= \langle A^*g, A^*f \rangle = \overline{\langle A^*f, A^*g \rangle} = \overline{\langle f, AA^*g \rangle} = \langle AA^*g, f \rangle, & (AA^*)^* &= AA^*. \end{aligned}$$

Alternatively, $(AA^*)^* = A^{**}A^* = AA^*$

Theorem 13 The operators a and a^* are adjoints of each other, $a^{**} = a$, and $a \circ a^*$ is self-adjoint.

Proof. It suffices to verify that $\langle \psi_\ell, a\psi_k \rangle = \langle a^*\psi_\ell, \psi_k \rangle$ for all $k, \ell \in \mathbb{Z}$.

To this end, recall that $a\psi_k(z) = \psi'_k(z) = k \cdot \frac{z^{k-1}}{\sqrt{k!}} = \frac{k}{\sqrt{k}} \cdot \frac{z^{k-1}}{\sqrt{(k-1)!}} = \sqrt{k} \cdot \psi_{k-1}$. Hence,

$$\langle \psi_\ell, a\psi_k \rangle = \langle \psi_\ell, \psi'_k \rangle = \sqrt{k} \cdot \langle \psi_\ell, \psi_{k-1} \rangle.$$

Thus, if $\ell \neq k-1$, then $\langle \psi_\ell, a\psi_k \rangle = 0$. In contrast, if $\ell = k-1$, then

$$\langle \psi_{k-1}, a\psi_k \rangle = \sqrt{k} \cdot \langle \psi_{k-1}, \psi_{k-1} \rangle = \sqrt{k}.$$

Similarly, recall that $(a^*\psi_\ell)(z) = z \cdot \psi_\ell(z) = \frac{z^{\ell+1}}{\sqrt{\ell!}} = \sqrt{\ell+1} \cdot \frac{z^{\ell+1}}{\sqrt{(\ell+1)!}} = \sqrt{\ell+1} \cdot \psi_{\ell+1}$. Hence,

$$\langle a^*\psi_\ell, \psi_k \rangle = \sqrt{\ell+1} \cdot \langle \psi_{\ell+1}, \psi_k \rangle.$$

Thus, if $\ell+1 \neq k$, equivalently, $\ell \neq k-1$, then $\langle a^*\psi_\ell, \psi_k \rangle = 0$. In contrast, if $\ell = k-1$, then

$$\langle a^*\psi_\ell, \psi_k \rangle = \sqrt{\ell+1} \cdot \langle \psi_{\ell+1}, \psi_k \rangle = \sqrt{k} \cdot \langle \psi_k, \psi_k \rangle = \sqrt{k}.$$

Therefore, $\langle \psi_\ell, a\psi_k \rangle = \langle a^*\psi_\ell, \psi_k \rangle$, for all $k, \ell \in \mathbb{Z}$, which shows that a^* is the adjoint of a .

Exercise 1 Find the eigenvalues and eigenfunctions of aa^* .

Verify that a^* and Q have no eigenfunctions in $\mathcal{H}$.

Are $P \circ P$ and $Q \circ Q$ self-adjoints?

1 Continuity

This material is a part of Functional Analysis [2], [3], [4].

Definition 14 A linear operator $A : \mathcal{H} \rightarrow \mathcal{H}$ is **bounded** if and only if there exists an upper bound $b \geq 0$ such that $\|Ah\| \leq b \cdot \|h\|$ for every $f \in \mathcal{H}$, equivalently, by linearity, $\|Af\| \leq b$ for every $f \in \mathcal{H}$ such that $\|f\| = 1$.

If A is bounded, then the subordinate **norm** of A , denoted by $\|A\|$, is the least such upper bound.

Theorem 15 A linear operator $A : \mathcal{H} \rightarrow \mathcal{H}$ is continuous if and only if it is continuous at 0.
A linear operator $A : \mathcal{H} \rightarrow \mathcal{H}$ is continuous if and only if it is bounded.

Proof. If A is continuous, then A is continuous at 0.

If A is continuous at 0, then for each $\varepsilon > 0$ there is a $\delta > 0$ such that for each $h \in \mathcal{H}$, if $\|h\| = \|h - 0\| < \delta$, then $\|Ah\| = \|Ah - 0\| < \varepsilon$. Hence, if $f \in \mathcal{H}$ and $g \in \mathcal{H}$ are such that $\|f - g\| < \delta$, then $\|Af - Ag\| = \|A(f - g)\| < \varepsilon$, which means that A is continuous at f .

Also, if $\|h\| = \frac{\varepsilon}{2} < \delta$, then $\|Ah\| < \varepsilon$. Thus, if $\|f\| = 1$, then set $h := \frac{\varepsilon}{2} \cdot f$, so that $\|h\| = \frac{\varepsilon}{2} < \delta$, and hence $\|Af\| = \|A \frac{2}{\varepsilon} \cdot h\| = \frac{2}{\varepsilon} \cdot \|Ah\| < \frac{2}{\varepsilon} \cdot \varepsilon$, which states that A is bounded.

If A is bounded, then $\|Ah\| \leq \|A\| \cdot \|h\|$ for every $h \in \mathcal{H}$. Thus, if $\|h\| < \delta := \frac{\varepsilon}{1+\|A\|}$, then $\|Ah\| \leq \|A\| \cdot \|h\| \leq \|A\| \cdot \frac{\varepsilon}{1+\|A\|} < \varepsilon$, which states that A is continuous at 0, and hence continuous everywhere. Allowing for equality and adding a positive constant may be necessary if $\|A\| = 0$.

Example 16 If $\omega > 0$, then the linear operator H is *not* continuous:

$$\|\psi_n\| = 1, \quad \lim_{n \rightarrow \infty} \|H\psi_n\| = \lim_{n \rightarrow \infty} \left\| \omega \cdot \left(n + \frac{1}{2}\right) \cdot \psi_n \right\| = \lim_{n \rightarrow \infty} \omega \cdot \left(n + \frac{1}{2}\right) = \infty.$$

Equivalently,

$$\varphi_n := \frac{1}{\sqrt{n}} \cdot \psi_n, \quad \|\varphi_n\| = \frac{1}{\sqrt{n}}, \quad \lim_{n \rightarrow \infty} \varphi_n = 0, \quad \lim_{n \rightarrow \infty} \|H\varphi_n\| = \lim_{n \rightarrow \infty} \omega \cdot \left(\sqrt{n} + \frac{1}{2\sqrt{n}}\right) = \infty.$$

In contrast, if $\omega > 0$, then the inverse operator, H^{-1} , is continuous, with $\|H^{-1}\| = \frac{2}{\omega}$:

$$\|\psi_n\| = 1, \quad \|H^{-1}\psi_n\| = \left\| \frac{1}{\omega \cdot \left(n + \frac{1}{2}\right)} \cdot \psi_n \right\| = \frac{1}{\omega \cdot \left(n + \frac{1}{2}\right)} \leq \frac{2}{\omega};$$

$$f = \sum_{n=0}^{\infty} c_n \cdot \psi_n, \quad \|H^{-1}f\|^2 = \|H^{-1} \sum_{n=0}^{\infty} c_n \cdot \psi_n\|^2 = \sum_{n=0}^{\infty} \frac{\|c_n\|^2}{\omega^2 \cdot \left(n + \frac{1}{2}\right)^2} \leq \frac{2^2}{\omega^2} \cdot \|f\|^2.$$

References

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